3 **(a)** A cylinder of fixed volume contains some nitrogen gas at a pressure of 2.1×10^5 Pa. The gas has a density 1.5 kg m^{-3} and the molar mass of nitrogen is 14 g, calculate

(i) the root-mean-square speed of the nitrogen molecules,

root-mean-square speed = …….m s^{-1} [2]

(ii) the temperature of the nitrogen gas in the cylinder.

temperature = …….K [2]

(iii) Using kinetic theory of gases, state and explain what will happen to the pressure of the nitrogen gas when the temperature of the gas is increased.

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……..…… [3]

(b) A refrigerator door is opened and room temperature air (25 °C) filled the 1.5 m^3 compartment of the refrigerator. A 2.0 kg chicken, also at room temperature, is placed in the refrigerator and the door is closed.

If the power rating of the refrigerator is 200 W, with an efficiency of 0.80, calculate the time required to cool the chicken and air in the refrigerator to thermal equilibrium at a temperature of 4.0°C.

(Density of air = 1.25 kg m^{-3}, specific heat capacity of air = 1.02×10^3 J kg^{-1} K^{-1}, specific heat capacity of chicken = 3.48×10^3 J kg^{-1} K^{-1})

time required = …….s [3]